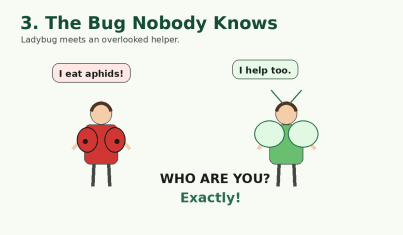


THE BUG NOBODY KNOWS

A 5-Minute FLL BIOGLOW Innovation Project Play

SEE IT. LEARN IT. PROTECT IT. SHARE IT.



CAST

CHILD	ROLE	MAIN PROP
Kid 1	Food Detective / Narrator	grocery basket + magnifying glass
Kid 2	Farmer / Gardener	straw hat + plant
Kid 3	Ladybug	red wings/headband
Kid 4	Green Lacewing	green wings + antennae
Kid 5	Young Engineer / Scientist	safety goggles + prototype

Competition reminder: the costumes are fun, but the winning story is the evidence of learning, testing, feedback, and improvement.

5-MINUTE STORY MAP

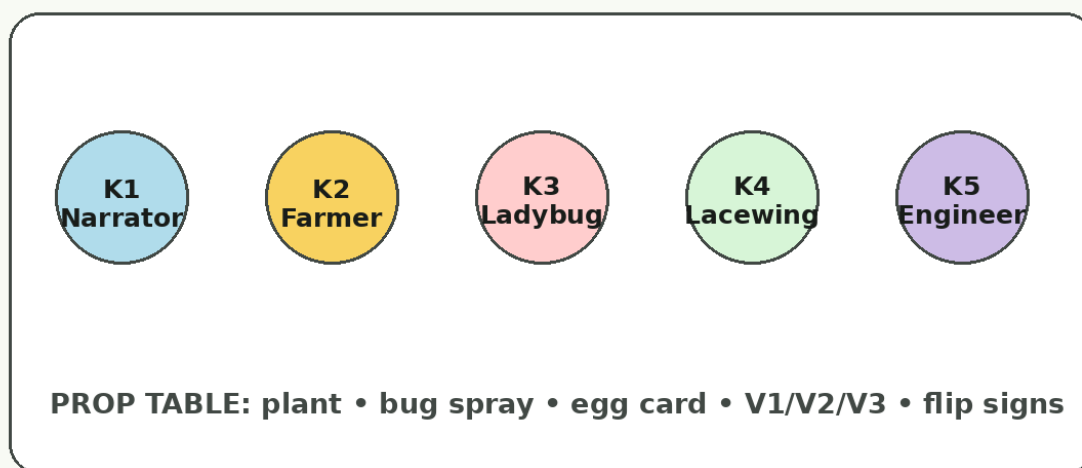
The play should feel fast, playful, and scientific. Do not rush the punch lines; do rush prop changes.

TIME	SCENE	JUDGE MUST UNDERSTAND	FLL STEP
0:00-0:35	1	Our food connects us to a farm ecosystem.	IDENTIFY
0:35-1:05	2	Farms face weeds, rabbits, and crop pests.	IDENTIFY
1:05-1:40	3	Green lacewings are useful but unfamiliar.	RESEARCH
1:40-2:35	4	The real problem: people may remove beneficial insects because they cannot recognize them.	PROBLEM
2:35-3:05	5	Current products are mainly for release, not kid observation.	DESIGN GAP
3:05-4:15	6	Kids build imperfect prototypes, test, get feedback, and improve.	CREATE / ITERATE
4:15-5:00	7	The solution teaches others and supports biodiversity.	COMMUNICATE

STAGE SETUP

Stage Setup

Keep movement simple so the science stays clear.



Keep all props on one side. No one should turn their back to the judges for more than a second.

SCENE 1 — WHERE DOES OUR FOOD COME FROM?

0:00–0:35



KID 1: We started BIOGLOW with a very important scientific question... Where did my carrot come from?

(Holds up carrot dramatically.)

KID 3: The grocery store!

KID 5: Costco!

KID 2: Seriously?!

(Everyone reacts.)

KID 1: Okay... before the grocery store!

KID 2: A farm!

KID 1: So we visited Coastal Roots Farm near our homes and started looking at a farm differently — not just as a place that grows food...

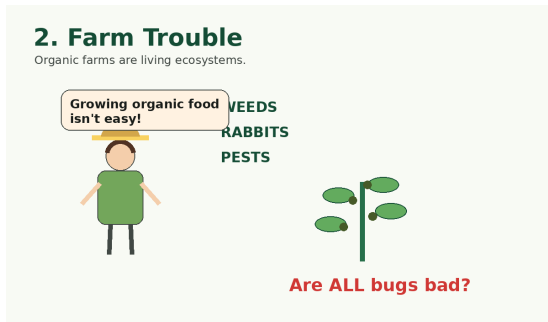
ALL: ...but as an ECOSYSTEM!

(Step forward together.)

DIRECTOR NOTE Pause after “Costco!” for the laugh. The science point is that food begins inside an ecosystem, not a store.

SCENE 2 — FARM TROUBLE

0:35–1:05



KID 2: Growing organic food isn't easy! Farmers deal with weeds... rabbits... and insects eating crops!

(Pull paper weed; make quick bunny ears; point to aphids.)

KID 1: So we asked: Are all bugs bad for farms?

KID 3: Did somebody say BUGS? Don't worry! I'm a ladybug! I eat pests like aphids.

(Ladybug enters superhero-style.)

KID 1: Everybody knew ladybugs. They're red, cute, and in books and toys.

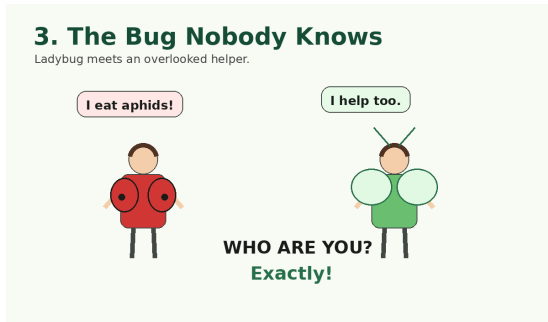
KID 5: But are ladybugs the ONLY insects helping farmers?

ALL: NO!

DIRECTOR NOTE Keep this scene brisk. It sets up the contrast between a famous beneficial insect and an overlooked one.

SCENE 3 — THE BUG NOBODY KNOWS

1:05–1:40



KID 4: Excuse me... I help too.

(Lacewing walks in quietly.)

ALL FOUR: Who are YOU?

(Turn toward Kid 4.)

KID 4: Exactly! I'm a green lacewing. My larvae eat aphids and other small crop pests, and farmers can use lacewings as part of natural pest control.

(Pause after "Exactly!")

KID 3: Wait. You help farms too? Why does everybody know ME... but almost nobody knows YOU?

KID 4: That became our question too.

DIRECTOR NOTE The "Who are YOU?" / "Exactly!" exchange is a memorable laugh line. Do not over-act it.

SCENE 4 — THE REAL-WORLD PROBLEM

1:40–2:35



KID 2: Oh no! Bugs! Bugs are eating my plant! I know what to do — SPRAY THEM ALL!

(Grabs empty spray bottle labeled BUG SPRAY.)

KID 3: WAIT! I'm a good bug! I eat aphids!

(Ladybug steps in front of plant.)

KID 4: Me too! I help control pests!

(Lacewing joins.)

KID 2: Wait... you mean not every bug is bad?

(Slowly lowers spray bottle.)

KID 1: That's when we realized our real problem was bigger than green lacewings. Many kids and families can't recognize beneficial insects.

KID 5: If we don't know which bugs are helping, we might remove the good bugs too — and depend more on pesticides that can also hurt biodiversity.

KID 2: So maybe before I spray... I should find out what the bug is!

(Puts spray bottle down.)

ALL: YES!

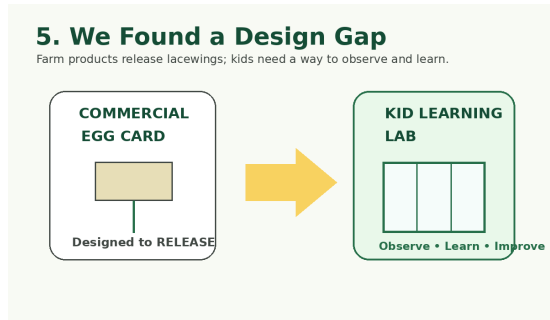
KID 5: Our question became: How might we help kids recognize, observe, and protect beneficial insects?

(Flip up PROBLEM sign.)

DIRECTOR NOTE This is the most important problem-definition scene. The judges should SEE the problem, not just hear a definition of biodiversity.

SCENE 5 — HOW CAN KIDS ACTUALLY LEARN?

2:35–3:05



KID 3: We searched for ways kids could learn by actually observing beneficial insects. Ladybugs? Easy — there are kits, toys, books, and habitats.

(Hold up ladybug kit/photo.)

KID 4: Green lacewings? Not so easy.

(Looks around.)

KID 5: We found commercial green lacewing egg cards designed mainly to release insects onto plants.

(Pretend to hang egg card on plant.)

KID 1: That works for farmers, but it wasn't designed for children who want to observe and learn from the life cycle.

KID 2: So we thought...

ALL: What if WE designed one?

DIRECTOR NOTE Important wording: do not claim no education product exists anywhere. Say you found a gap in the products available to you and in common kid learning tools.

SCENE 6 — THE LACEWING LEARNING LAB

3:05–4:15

KID 5: Introducing... the Lacewing Learning Lab!

(Bring out rough V1.)

ALL: Ta-da!

KID 5: Our first idea was simple: put them in a clear box and watch them grow.

KID 4: Ummm... there are a few problems with that.

KID 5: We learned lacewing larvae can even attack each other.

KID 3: They need ventilation.

KID 1: We need a safe way to feed and observe them.

KID 2: And tiny larvae can escape.

KID 4: If we want to observe the life cycle, we also need space for cocoons and adults.

KID 5: So the adults did NOT give us the final answer. We ordered eggs, built different ideas, and tested visibility, ventilation, feeding, escape prevention, and space to grow.

(Flip TEST → IMPROVE.)

KID 5: Our first design wasn't perfect. GOOD! Because that's how engineers learn.

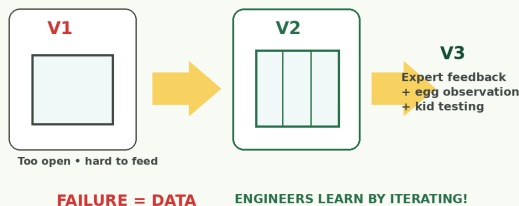
(Hold up V1, then V2/V3.)

KID 2: At Coastal Roots Farm, we learned: [ADD ONE REAL FARMER FEEDBACK].

KID 5: So we changed: [ADD THE REAL DESIGN CHANGE].

6. Build • Test • Improve

The first design should NOT be perfect.



DIRECTOR NOTE Never fake the brackets. After the farm tour, replace them with one concrete quote/idea and one visible change in the prototype.

SCENE 7 — BIGGER THAN A BOX + FINAL CALLBACK

4:15–5:00

KID 1: Then we realized something: our solution isn't really just a bug box.

(Flip SHARE sign.)

KID 4: The box helps kids SEE a beneficial insect.

KID 3: Our activities help them LEARN what it does.

KID 2: Our garden guide helps families PROTECT beneficial insects.

KID 5: And our classrooms, schools, community events, farmers markets, and website help us SHARE what we learned.

(Point to impact poster.)

KID 1: We tested our learning activity with [NUMBER] kids. Before, [__%] could identify a green lacewing. After, [__%] could identify it and explain why beneficial insects matter.

(Only use real survey results.)

KID 2: So... should I be afraid of every bug in my garden?

(Holds plant.)

KID 3: No!

KID 4: First... get to know us!

KID 1: Because you can't protect something...

KID 5: ...if you don't know it's there.

ALL: SEE IT. LEARN IT. PROTECT IT. SHARE IT.

(Step forward.)

KID 4: And next time you see a bug, don't just ask, "Is it bad?"

ALL: ASK: "WHAT JOB DOES IT DO?"

(Freeze in final pose.)

7. Bigger Than a Box

One model insect can open the door to biodiversity.



Don't just ask: "Is it bad?"

Ask: "WHAT JOB DOES IT DO?"

DIRECTOR NOTE Bring back the "before you spray" idea visually by having the Farmer set the spray bottle far away from the plant during the final pose.

AFTER THE FARM TOUR: TURN THE PLAY INTO A WINNING EVIDENCE STORY

Do not fill these blanks with guesses. Let the kids collect the evidence and then update the script.

1. FARM EXPERT FEEDBACK

Write one thing the farmer/expert said that surprised the kids.

Judging use: “We learned that...”

2. V1 PROBLEM

What actually failed? Escape? Feeding? Visibility? Space?

Judging use: Show the imperfect V1 to the judges.

3. DESIGN CHANGE

What changed because of the feedback or observation?

Judging use: “So we changed...”

4. EGG OBSERVATION

What happened after the lacewing eggs arrived?

Judging use: Add 1 photo or simple observation chart.

5. KID IMPACT

How many kids did you teach? What changed before vs. after?

Judging use: Use a 2–4 question before/after survey.

6. NEXT USER

Who else could use this? Teacher? school garden? family? farmer market?

Judging use: Name a real next audience.

PROPS, COSTUMES & REHEARSAL CHECKLIST

KEEP COSTUMES SIMPLE

- Kid 1: magnifying glass / grocery basket
- Kid 2: straw hat
- Kid 3: ladybug headband or small red wings
- Kid 4: light green wings / antennae
- Kid 5: safety goggles

MUST-HAVE PROPS

- Carrot or vegetable
- Plant with removable paper aphids
- Empty bottle clearly labeled BUG SPRAY
- Commercial lacewing egg card/photo
- V1 rough prototype
- V2/V3 improved prototype or photos
- Flip cards: PROBLEM / TEST / IMPROVE / SHARE
- Impact poster: 1 Box → 5 Kids → Classrooms → Community → World

REHEARSAL RULES

- Aim for 4:40–4:50 in practice so there is room for laughter and movement.
- Every child must understand the science behind every line, not only memorize it.
- Practice with the real props from the beginning.
- One child always speaks while the others freeze or support the action.
- If a prop fails, keep going. Judges care more about understanding than theater perfection.

FINAL MESSAGE TO THE KIDS

Your first idea does NOT need to work. Your job is to notice, ask, test, learn, change, and explain WHY.